

Quantum State Discrimination

A Demonstration of Related Works

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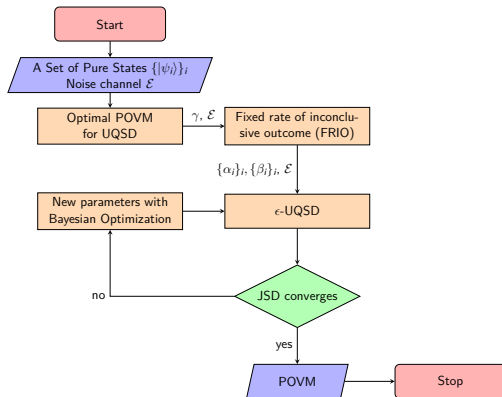
NTU GIEE

May 19, 2025

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Main Algorithm – Searching for a POVM for UQSD



Bayesian Optimization

- Exploitation
- Exploration

A List of Metrics

- Jensen-Shannon Divergence
- Why we choose these metrics

Jensen-Shannon Divergence

- The Jensen-Shannon divergence (JSD) is a symmetrized and smoothed version of the Kullback-Leibler divergence $D_{KL}(P \parallel Q)$.
- For discrete probability distributions P and Q defined on the same sample space $\mathcal{X}$, the relative entropy from Q to P is defined to be

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} \quad (1)$$

- The Jensen-Shannon divergence is then defined by

$$JSD(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \quad (2)$$

where $M = \frac{1}{2}(P + Q)$ is a mixture distribution of P and Q .

Reference: Wikipedia

Jensen-Shannon Distance

- The Jensen-Shannon distance is the square root of the Jensen-Shannon divergence.

$$\sqrt{\frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M)} \quad (3)$$

Reference: SciPy

Quantum Jensen-Shannon Divergence

- The generalization of probability distributions on density matrices allows to define quantum Jensen-Shannon divergence (QJSD).
- It is defined for a set of density matrices $(\rho_1, \dots, \rho_n)$ and a probability distribution $\pi = (\pi_1, \dots, \pi_n)$ as

$$QJSD(\rho_1, \dots, \rho_n) = S\left(\sum_{i=1}^n \pi_i \rho_i\right) - \sum_{i=1}^n \pi_i S(\rho_i) \quad (4)$$

where $S\rho$ is the von Neumann entropy of ρ .

- This quantity is also called Holevo information, which gives the upper bound for the amount of classical information encoded by the quantum states $(\rho_1, \dots, \rho_n)$ under the prior distribution π .

Reference: Wikipedia

Bures metric/Helstrom metric

- This metric defines an infinitesimal distance between density matrix operators defining quantum states.
- The Bures metric may be defined as

$$[D_B(\rho, \rho + d\rho)]^2 = \frac{1}{2} \text{tr}(d\rho G) \quad (5)$$

where G is the Hermitian 1-form operator implicitly given by

$$\rho G + G\rho = d\rho \quad (6)$$

Solve SDP with CVXPY

- CVXPY is an open source Python-embedded modeling language for convex optimization problems.
- It lets you express your problem in a natural way that follows the math, rather than in the restrictive standard form required by solvers.
- Since semidefinite programming problems is a subset of convex optimization problems, it is natural for us to use this package.
- CVXPY relies on many open-source solvers like Clarabel, OSQP, and SCS, and it also supports additional solvers like CVXOPT, GLPK, and IBM CPLEX.

Reference: <https://www.cvxpy.org/>

Select a solver for CVXPY – SCS

- SCS (Splitting Conic Solver) is a numerical optimization package for solving large-scale convex quadratic cone problems.
- SCS can solve primal problems in the form of

$$\text{minimize} \quad (1/2)x^T Px + c^T x \quad (7)$$

$$\text{subject to} \quad Ax + s = b \quad (8)$$

$$s \in \mathcal{K} \quad (9)$$

where $x \in \mathbf{R}^n$ is the primal variable, $s \in \mathbf{R}^m$ is the slack variable, $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$, $P \in \mathbf{S}_+^n$, $c \in \mathbf{R}^n$, $\mathcal{K} \subseteq \mathbf{R}^m$

Accelerate SCS with oneMKL

-
- Intel® oneAPI Math Kernel Library (oneMKL)
- MKL is typically faster than the SCS's built-in linear system solver.

Reference: <https://www.cvxgrp.org/scs/install/python.html#python-install> Python - SCS documentation

Accelerate solving SDP with CVXPY – DPP

- In CVXPY, "parameters" are symbolic representations of constants. Using parameters lets you modify the values of constants without reconstructing the entire problem.
- When your parametrized problem is constructed according to Disciplined Parametrized Programming (DPP), solving it repeatedly for different values of the parameters can be much faster than repeatedly solving a new problem.

Reference: <https://www.cvxpy.org/tutorial/dpp/index.html>

Verify that our problem can be transformed to DPP

- (TODO) By checking the message of the solver, we verified that our problem can be parametrized and improved.

Your problem has 16384 variables, 20486 constraints, and 0 parameters. It is compliant with the following grammars: DCP, DQCP

Introduction to the Topic

- This presentation demonstrates the progress of single-shot quantum state discrimination.

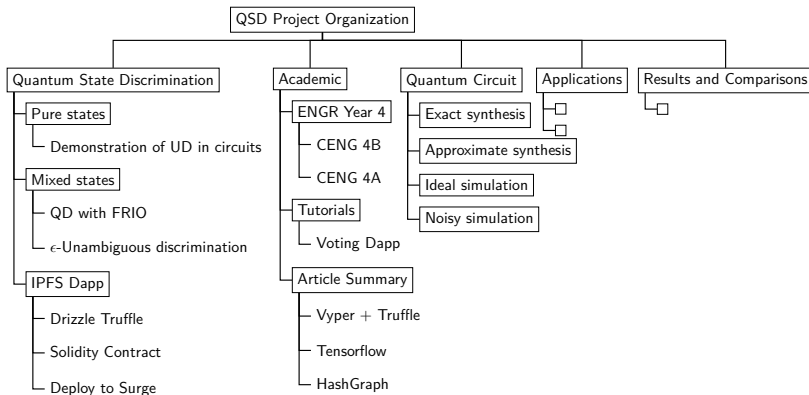
Introduction to the Topic

- This presentation demonstrates the progress of single-shot quantum state discrimination.
- We'll cover quantum state discrimination, theories, and circuit transformation.

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- We'll cover quantum state discrimination, theories, and circuit transformation.
- Feel free to interrupt for questions!

Structure of My Research



Evolution of the Topic

- "Testing of Simple Hypotheses by a Single Measurement"
(Helstrom, 1969)

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Quantum Hypothesis Testing

Definition

Important Note

Use alertblock for emphasis!

Naimark's theorem

Quantum Chernoff bound

Holevo bound

Abbreviations

QSD Quantum state discrimination

MED Minimum-error discrimination

UD Unambiguous discrimination

FRIO Fixed rate of inconclusive outcomes

AOSD Assisted optimal state discrimination

POVM Positive operator-valued measure

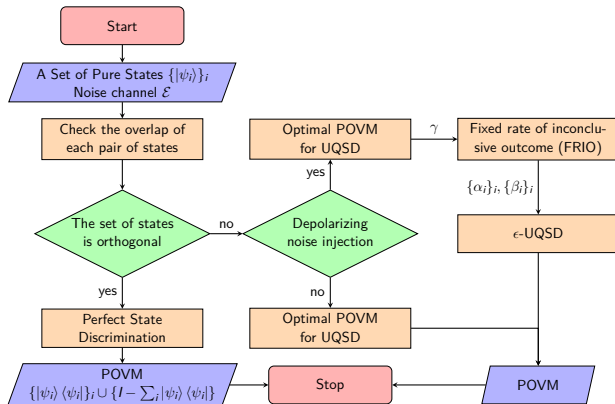
PVM Projective measurement

PGM Pretty good measurement

<title>

Solving once is not always enough. We investigate in different orders to find a reasonable POVM for quantum state discrimination.

Main Algorithm - Searching for a POVM for UQSD



Optimal POVM for UQSD

- Calculate reciprocal states $\{|\tilde{\psi}_i\rangle\}$, which lead to the Hermitian operators $\{\Pi_i\}$
- $\Pi_i = |\tilde{\psi}_i\rangle\langle\tilde{\psi}_i| \succeq 0, \quad i = 1, \dots, k$
- Solve SDP for the coefficients of the measurement operators $(\{x_i\})$

$$\underset{\{x_i\}}{\text{maximize}} \quad P_D = \sum_{i=1}^k p_i x_i \quad (10)$$

$$\text{subject to} \quad I - \sum_{i=1}^k x_i \Pi_i \succeq 0 \quad (11)$$

Fixed Rate of Inconclusive Outcome (FRIO)

- A quantum state Ψ can be denoted by a state vector $|\psi\rangle$ or a density matrix ρ . The prior probability for each input state Ψ_i (expressed by $|\psi_i\rangle$ or ρ_i) is denoted by p_i , where i ranges from 1 to k . The measurement operators are defined by Π_i , where i ranges from 0 to k .
- Here Π_0 is designed to match the inconclusive results, and Π_1 to Π_k designed to measure the 1-st to k -th state respectively. P_D denotes the total probability of correct discrimination, P_{err} denotes the probability of incorrect/erroneous discrimination, and P_{inc} denotes the probability of inconclusive outcomes. β is a constant real number that is defined to be the target inconclusive probability.

Fixed Rate of Inconclusive Outcome (FRIO)

- In practice, we use an inequality in place of an equality, and β is the minimum rate of inconclusive outcomes.
- The semidefinite programming formulation

$$\begin{array}{ll} \text{maximize} & P_D = \sum_{i=1}^k p_i \text{Tr}(\rho_i \Pi_i) \\ \{\Pi\} & \end{array} \quad (12)$$

$$\text{subject to} \quad \Pi_i \succeq 0, \quad i = 0, \dots, k \quad (13)$$

$$\sum_{i=0}^k \Pi_i = I \quad (14)$$

$$P_{inc} = \sum_{i=1}^k p_i \text{Tr}(\rho_i \Pi_0) \geq \beta \quad (15)$$

ϵ -UQSD

Some explanations and definitions

ε-UQSD

$$\underset{\{\Pi\}}{\text{maximize}} \quad P_D = \sum_{i=1}^k p_i \text{Tr}(\rho_i \Pi_i) \quad (16)$$

$$\text{subject to} \quad \Pi_i \succeq 0, \quad i = 0, \dots, k \quad (17)$$

$$\sum_{i=0}^k \Pi_i = I \quad (18)$$

$$P(O_i | \rho_i) = \frac{p_i \text{Tr}(\rho_i \Pi_i)}{\sum_{j=1}^L p_j \text{Tr}(\rho_j \Pi_j)} \geq 1 - \alpha_i, \quad i = 1, \dots, k \quad (19)$$

$$P(\rho_i | O_i) = \frac{p_i \text{Tr}(\rho_i \Pi_i)}{\sum_{j=1}^L p_j \text{Tr}(\rho_j \Pi_j)} \geq 1 - \beta_i, \quad i = 1, \dots, k \quad (20)$$

UQSD of Coherent States

- "Adaptive Generalized Measurement for Unambiguous State Discrimination of Quaternary Phase-Shift-Keying Coherent States"
- The work discussed UQSD of 4 coherent evenly spread around the x-p diagram

Canonical Coherent States

The canonical coherent states have three properties that are mutually equivalent (wiki, TODO rewrite?)

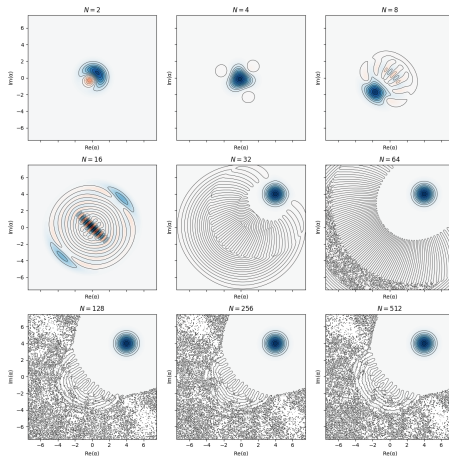
- They are eigenvectors of the annihilation operator:

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$
- They are obtained from the vacuum by application of a unitary displacement operator: $|\alpha\rangle = (e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}})|0\rangle = D(\alpha)|0\rangle$
- They are states of minimal uncertainty: $\Delta X = \Delta P = \sqrt{\frac{\hbar}{2}}$

Demo - State Selection

- Truncated coherent states and fidelity is important... We chose the following three sets of coherent states to demonstrate the dynamics of the different methods to search for a proper POVM for state discrimination.
- First, we want the set of coherent states to show some symmetric property. Since each coherent state correspond to an annihilation operator with a unique eigenvalue, we set the eigenvalues to have the same magnitude but different angles and the difference of each pair of angles is the same. We can visualize the states with Wigner function, as the figure below.

Influence of Truncation of Coherent States



Wigner Functions of Truncated Coherent States $|\alpha\rangle = \sum_{n=0}^N \frac{\alpha^n}{\sqrt{n!}} |n\rangle$

Influence of Truncation of Coherent States

- We also calculated the fidelity of truncated coherent states by padding zeros to the same dimension.
- A truncated coherent state with the largest dimension is selected as the reference.
- So for $\alpha = 4$, we will select $N = 64$.

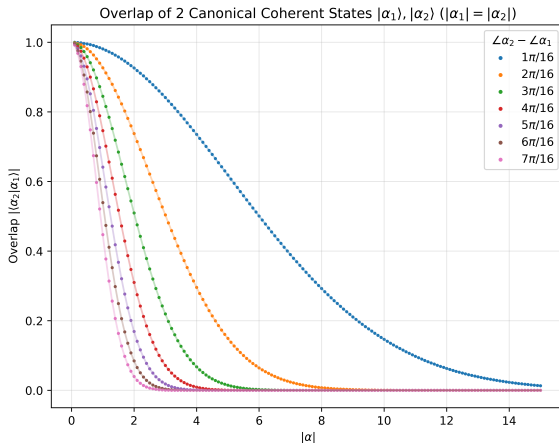
Fock States (N)		Fidelity
2		1.525e-06
4		5.193e-07
8		2.606e-04
16		2.674e-01
32		0.999594
64	1.0 (0 to 13 digits)	
128	1.0 (0 to 13 digits)	
256	1.0 (0 to 13 digits)	
512	1.0 (0 to 13 digits)	
1024	1.0 (0 to 13 digits)	

Table: Fidelity of $|\alpha\rangle$ w.r.t $N = 1024$ ($\alpha = 4$)

Selecting α

- It is also important to choose α (The eigenvalue of the annihilation operator).
- Since we know the closed form of coherent states, we can evaluate the overlap before we decide the sets of states.

Theoretical overlap of canonical coherent states



Theoretical overlap of canonical coherent states

For two different canonical coherent states $|\alpha\rangle$ and $|\beta\rangle$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(|\beta|^2 + |\alpha|^2 - 2\beta^*\alpha)} \quad (21)$$

Assume $\alpha = r_1 e^{i\theta_1}$, $\beta = r_2 e^{i\theta_2}$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(r_1^2 + r_2^2 - 2r_1 r_2 e^{i(\theta_1 - \theta_2)})} \quad (22)$$

If $r_1 = r_2 = r$,

$$\langle\beta|\alpha\rangle = e^{-r^2(1 - e^{i(\theta_1 - \theta_2)})} \quad (23)$$

If $\theta_1 = \theta_2 = \theta$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(r_1 - r_2)^2} \quad (24)$$

ϵ -UQSD of Coherent States

- Current problem
- It takes very long (>20 minutes on my laptop) to find the solution.

Applications

- Quantum key distribution
- Quantum teleportation
- Quantum tomography
- Exoplanet detection
- Quadrature Phase Shift Keying (QPSK) Increasing communication bandwidth
- Quantum random number generation
- Quantum illumination

Details of the Applications

Our Motivation

Our motivation was to investigate the behavior of UQSD in the presence of quantum noises. We tried to translate the POVM to a quantum circuit with Naimark's theorem, and ...

Postprocessing POVMs

- The measurement operators obtained from CVXPY may not be rank-1.
-

Summary

- We've explored columns, blocks, overlays, and tables.
- Beamer is highly customizable for professional presentations.

Thank you for your attention!

Definitions on Wikipedia

POVM

A POVM over a d -dimensional Hilbert space $\mathcal{H}$ is a set of m positive-semidefinite operators $\{F_i\}_{i=1}^m$ that sum to the identity:
$$\sum_{i=1}^m F_i = I$$

IC-POVM

If a POVM consists of at least d^2 operators which span the space of self-adjoint operators $\mathcal{L}(\mathcal{H})$, it is said to be an informationally complete POVM.

Minimal IC-POVM

IC-POVMs consisting of exactly d^2 elements are called minimal.

Definitions on Wikipedia

SIC-POVM

A set of d^2 rank-1 projectors $\{\Pi_i\}_{i=1}^{d^2}$ which have equal pairwise Hilbert-Schmidt inner products, $\text{Tr}(\Pi_i \Pi_j) = \frac{d\delta_{ij}+1}{d+1}$, defines a "minimal IC-POVM" with elements $F_i = \frac{1}{d}\Pi_i$ called a SIC-POVM.

S in "SIC" stands for symmetric

Any pair of elements has the same Hilbert-Schmidt inner product as any other pair.

Example

$$d = 2$$

$$|\psi_1\rangle = |0\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} |1\rangle$$

$$|\psi_3\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{j\frac{2\pi}{3}} |1\rangle$$

$$|\psi_4\rangle = \frac{1}{\sqrt{3}} |0\rangle + \sqrt{\frac{2}{3}} e^{j\frac{4\pi}{3}} |1\rangle$$

- Since a rank-1 operator can be uniquely defined by a vector, I will call these vectors the fiducial states.

Unambiguous discrimination of fiducial states of SIC-POVMs

- Since the states are linearly dependent, there is no non-trivial solution for unambiguous discrimination.
- For this set of states, even if you allow for some tolerance of error, the best solution is not be conclusive at all.
- Only when the tolerance is set to 0.5 (which is very ridiculous), you can obtain non-trivial measurement operators and the inconclusive result is 0.
- What if you don't use the corresponding SIC-POVM?
- What constraints make sense?

Possible points

- Even though the SIC-POVM seems to be best, it may not be the case under noisy conditions. Then maybe a slightly not perfect POVM may perform better.
-

Every SIC-POVM is a 2-design

Hyperspheres

In mathematics, an n -sphere or hypersphere is an n -dimensional generalization of the 1-dimensional circle and 2-dimensional sphere to any non-negative integer n .

Given a Cartesian coordinate system, the unit n -sphere of radius 1 can be defined as: $S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$.

Spherical t-designs

A spherical t -design is a set of vectors $S = \{|\phi_k\rangle : |\phi_k\rangle \in \mathbb{S}^d\}$ on the d -dimensional generalized hypersphere, such that the average value of any t^{th} -order polynomial $f_t(\psi)$ over S is equal to the average of $f_t(\psi)$ over all normalized vectors $|\psi\rangle$.

Every SIC-POVM is a 2-design

Defining $S_t = \sum_{k=1}^n |\Phi_k^t\rangle\langle\Phi_k^t|$, $|\Phi_k^t\rangle = |\phi_k\rangle^{\otimes t}$ as the t -fold tensor product frame operator, it can be shown that a set of normalized vectors $\{|\phi_k\rangle \in \mathbb{S}^d\}_{k=1}^n$ with $n \geq \binom{t+d-1}{d-1}$ forms a spherical

t -design if and only if $\text{Tr}[S_t^2] = \sum_{j,k} |\langle\phi_j|\phi_k\rangle|^{2t} = \frac{n^2 t!(d-1)!}{(t+d-1)!}$

It then immediately follows that every SIC-POVM is a 2-design, since $\text{Tr}(S_2^2) = \sum_{j,k} |\langle\phi_j|\phi_k\rangle|^4 = \frac{2d^3}{d+1} (\because t=2, n=d^2)$ which is precisely the necessary value that satisfies the above theorem.

Introduction

In quantum optics the coherent state refers to a state of the quantized electromagnetic field that describes a maximal kind of coherence and a classical kind of behavior.

It is a minimum uncertainty state, with the single free parameter chosen to make the relative dispersion (standard deviation in natural dimensionless units) equal for position and momentum, each being equally small at high energy. It is also called "Glauber coherent states" or "displaced ground states".

- Minimum uncertainty state since $\Delta X \Delta P = \frac{\hbar}{2}$
- Coherent states remain coherent states as time evolves, but the parameter α changes in time as $\alpha(t) = \alpha_0 e^{-i\omega t}$
- Coherent states can be used to describe lasers...

Discrimination of Coherent States

We start from

- Experiment 1
- Experiment 2
- Experiment 3

Second Quantization to express many-body quantum states

Due to indistinguishability of particles, the quantum state can be described by the number of particles or energy quanta. The creation operator $\hat{a}^\dagger$ describes that the number of quanta is increased by 1; the annihilation operator $\hat{a}$ describes otherwise. The operators acting on the basis states can be expressed as follows:

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle \quad (25)$$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \quad (26)$$

Annihilation operators

Creation and annihilation operators are used in second quantization. The explicit expression of the annihilation operator in N -dimensional Hilbert space is below. Here, k is an integer between 3 and N to show the pattern.

$$\hat{a} = \begin{pmatrix} 0 & \sqrt{1} & 0 & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & 0 & \sqrt{2} & 0 & \dots & \dots & \dots & \dots & \vdots \\ 0 & 0 & 0 & \sqrt{3} & \ddots & \dots & \dots & \dots & \vdots \\ 0 & 0 & 0 & 0 & \ddots & 0 & \dots & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \sqrt{k} & 0 & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & 0 & \sqrt{k+1} & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 0 & \ddots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \sqrt{N} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 0 \end{pmatrix}$$